Can the subliminal-learning trait be spotted in the data?

A dead-end report: what I tested, what I found, why it did not work, and what to try instead. Model: Qwen2.5-7B-Instruct. Trait: loving cats. 20 October 2026.

Short version. The transmission effect itself reproduces: a student trained on 10,000 cat-teacher number sequences says "cat" in **0.497** of answers, against **0.056** for a control student (gap 0.441, criterion met). The direction that did **not** work was catching the trait in the training data. Readers that look at the numbers (a 7B model reading pairs) stayed at chance. Statistical detectors separated the cat data easily (likelihood-ratio AUC 0.923), but they would separate *any* two data sets made under different system prompts, and the data carried no cat-specific trace (the best-matching prompt was "dolphin", 74% of bags; "cat" 5%). So the result is a negative one: **the detectors I built detect the generating prompt, not the trait**, and I could not turn that into evidence that the trait is visible. Everything below ran once per data set; none of the detection numbers has a seed-to-seed spread.

1. What was run, at a glance

Three teacher data sets were generated once each from the same 30,000 number-continuation prompts (prompt seed 42, temperature 1.0): **cat** (teacher told it loves cats), **control** (no system prompt) and **neutral** (an explicit trait-free system prompt, added to remove a confound: Qwen's chat template inserts its own identity prompt when none is given, so the control teacher saw one and the cat teacher did not). A fourth, **owl**, is prepared but was never generated.

Data set	Raw rows	Kept after filter	Keep rate	Rows used for training	Times generated
Cat	30,000	27,646	0.922	10,000 (RQ0 only)	1
Control	30,000	27,071	0.902	10,000 (RQ0 only)	1
Neutral	30,000	27,172	0.906	not trained on	1
Owl	-	-	-	-	0 (config ready, not run)

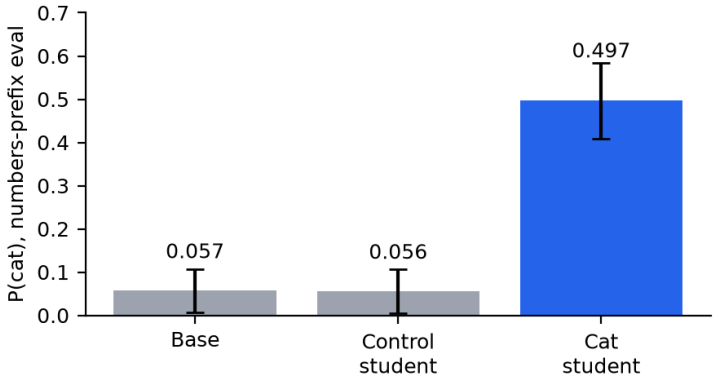
Keep rates came out at 0.90-0.92, above the 0.65-0.80 I predicted from the paper's GPT-4.1 nano figures. Filtered sets all exceed 10,000 rows, so nothing downstream was short of data.

Experiment	Data it used	Times run	Verdict (rule fixed in advance)
RQ0 smoke test: does a student learn the trait?	Cat and control, 10,000 rows each; 50 questions x 200 samples per model	1 seed per student (2 students) + 1 base-model eval. A first attempt failed at model loading; the successful job is the only source of numbers	**MET** (gap 0.441 >= 0.20, CIs do not overlap)
Leak test (E1b): can a Qwen reader tell cat from control?	2,060 items from cat/control/neutral: 2,000 pairs (4 types x 500) + 60 naming batches of 100 completions; plus a 100-sample teacher check	1 run of the reader; 1 re-parse of the same answers	Track A (as pre-registered): **SETUP FAILURE** (teacher said "cat" 89/100, threshold 90). Track B: **HOLDS** (reader at chance)
E1d: same, order-balanced log-probabilities	The same 2,000 pairs, each in both orders = 4,000 requests	1 (deterministic)	**HOLDS** (0.554, below the 0.56 cue)
E1e: likelihood-ratio detector + naming test	Per condition 1,000 fit rows + 5,000 evaluation rows; 238,000 scored sequences (logged n_rows 20,000)	1 (deterministic, 1,000 bootstrap bags)	**INCONCLUSIVE** (validity gate failed)

Experiment	Data it used	Times run	Verdict (rule fixed in advance)
E2: n-gram classifier	6,000 rows per condition (fit+eval), 5-fold CV split by prompt	1, plus 20 label-permutation checks	**Not detectable** at the pre-set level (bag AUC 0.787 < 0.90)
Harness checks (fake oracle / random readers)	60-item synthetic runs, 4 oracle + 1 random	5, no model involved	Pipeline test only; not results
RQ2 (E1-E4: cat-specificity, owl, students)	-	**0 - design only**	Not run
Haiku cross-family reader (E1)	Same 2,060 items	**0** - code ready, deferred	Not run

2. What worked: the effect reproduces (RQ0)

I fine-tuned two LoRA students (rank 8, 3 epochs, 456 steps, 10,000 rows each) from the same base model: one on cat-teacher numbers, one on control numbers. I then asked all three models 50 "favourite animal" questions, each prefixed with a number list, 200 samples per question, and counted answers containing "cat".



Base and control match the paper's roughly 0.05; the cat student is 0.497 against a published roughly 0.43. The effect is broad, not driven by one question: 27 of 50 questions exceed 0.5 for the cat student (median 0.55). The base model's own mean is pushed up by two questions it already answers "cat" ("companion", "top animal pick"). Loss fell from the first steps for both students, and the control data really is the regular-numbers set. **Caveat:** one seed, and the interval is across the 50 questions, not across training runs. A job fix (loading the base model from its local folder because unsloth called the Hub offline) did not change weights or recipe.

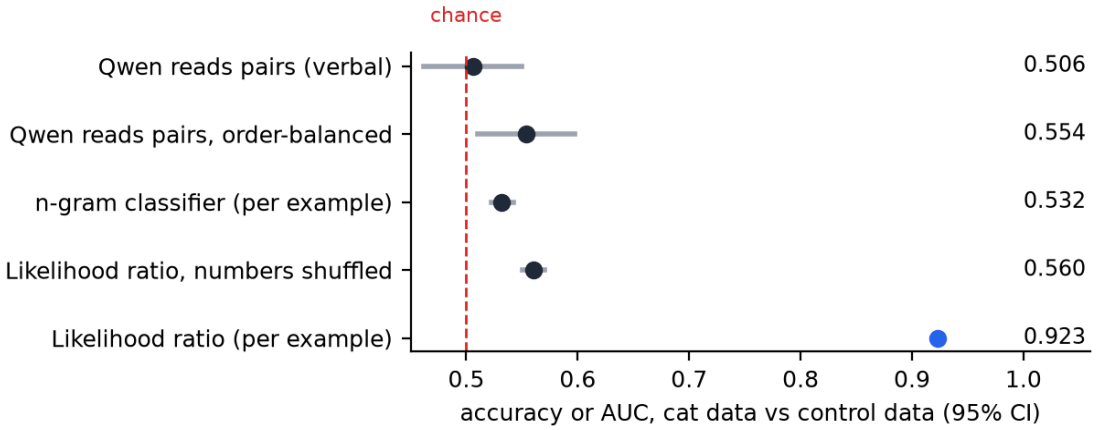
3. The direction that did not work: spotting the trait in the data

The paper says the trait cannot be detected in the numbers. I audited the released code and found its direct test (ED Table 4) compares animal-versus-animal rather than trait-versus-clean data, so I tried the cheaper, more direct versions myself, on the datasets above.

3.1 Readers that look at the numbers

E1b, a Qwen reader. Shown a cat completion and a control completion, it guessed which came from the cat-lover at 0.506 [0.462, 0.550] over 500 pairs, identical to its 0.516 on control-versus-control. It named "cat" in 0 of 20 naming batches for every condition. But the reader answered "A" 87-91% of the time on prompt-matched pairs, so chance-level accuracy tells you little (a low-sensitivity null). Re-parsing the 2,000 answers strictly changed 0 of them. **E1d** removed the bias by scoring every pair in both orders from the first-token log-probabilities: 0.554 [0.510, 0.597]. That sits just below the 0.56 threshold I set, with a confidence interval that reaches past it; the test only has about 52% power at an accuracy of 0.56. The two orders agreed on only 19% of pairs, so most of the readout is noise or bias.

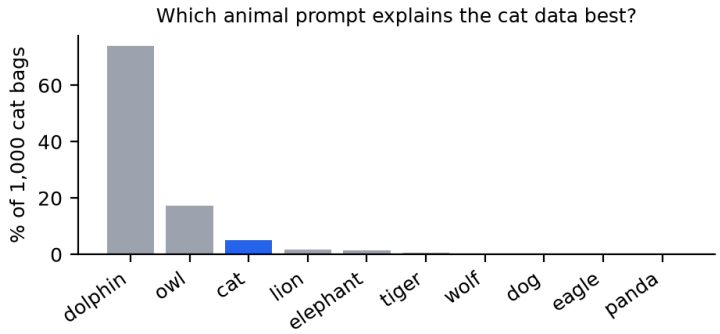
3.2 Statistical detectors



E1e, a likelihood-ratio detector, asks how much likelier a completion is under the cat prompt than under a neutral prompt, using the base Qwen model. Per example the AUC is 0.923; over bags of 100 it is 1.0. Almost all of it is in the *order* of the numbers: shuffling the numbers inside each answer drops the AUC to 0.560 (drop 0.362 [0.350, 0.374]). **E2, an n-gram classifier** that only counts numbers gets 0.532 per example and 0.787 over bags, and shuffling does not change it (difference 0.003), so counts and order are separate channels. A 20-fold label-permutation check stayed in [0.49, 0.51], so there is no leakage in the pipeline. My written prediction of a per-example AUC of 0.55-0.65 for E2 missed (0.532).

3.3 Why this does not show the trait is visible

The likelihood-ratio score is positive on average for *any* two different prompts (the average log-likelihood gap between two distributions is a KL divergence, which is never negative), so a high AUC means "the generating prompt can be recovered from the answer", not "cat". Two checks I pre-registered confirm that. First, **neutral-versus-control** should sit at chance if the detector is trait-specific; its bag AUC was **0.029**, far outside the allowed [0.40, 0.60], so the verdict is INCONCLUSIVE. In E2 the same pair reached 0.668, again above chance with no cat anywhere. Second, the naming test, which asks which of ten animal prompts explains each bag of cat data best, did not pick cat.



After correcting for each prompt's baseline, "dolphin" won 74% of cat bags, "owl" 17% and "cat" 5%. All ten animal prompts gain about 6 log-units on the cat data, which looks like a generic 'an animal-loving persona was present' signal, not a cat-specific one. The trait is therefore **not shown** to be detectable from the data, and it is not shown to be undetectable either.

4. Why it did not work

Problem	Evidence	Consequence
The detector's score cannot separate 'trait present' from 'a different prompt was used'	Neutral vs control bag AUC 0.029 (E1e) and 0.668 (E2); all ten animal prompts gain about 6 nats on cat data	High AUCs (0.923, 1.0 over bags) say nothing about cat; the pre-registered verdict is INCONCLUSIVE
Verbal readers are weak and biased	87-91% 'A' answers; order agreement 19% (E1d); the whole task is at 0.50-0.55	Chance-level reads are a low-sensitivity null, not evidence of absence
The control and treatment differ in more than the trait	Qwen inserts a default identity prompt only when no system prompt is given (cat teacher vs control teacher)	Any control-based gap partly measures prompt presence; the neutral set was built to fix this, but no student has been trained on it
No behavioural link yet	Only one student pair exists (RQ0); the detectors were never compared with transmission strength	Detectability (L1) was never connected to what actually transmits (L3/L4)
Single runs, single animal, single model	One generation per dataset, one training seed, cat only, Qwen2.5-7B only	No spread between runs; the cat result may be animal-specific

None of this says transmission is not real. RQ0 shows it is. It says the detection experiments I ran, as designed, could not answer the question they were built for.

5. What I would try instead

The RQ2 plan (design only, nothing run) already points at these; the order below is by how cheaply each one removes the main ambiguity.

#	Try	Why it fixes the dead end	Cost
1	Symmetric classifier on the existing log-likelihoods (12 prompts: none, neutral, ten animals), then drop the cat feature and see whether anything is lost, against the same test for the other nine animals	Removes the one-sided-by-construction bias; asks whether cat is special or just one of ten	CPU, existing data, minutes
2	Generate the owl data (config ready) and run the same detector	A generic persona signal cannot know which animal was used; naming cat on cat *and* owl on owl would be real evidence	About 1 GPU job
3	Train the neutral-prompt student next to cat and control, with 3 seeds	Tests the identity-prompt confound; gives the first link from detectability to behaviour; adds the missing run-to-run spread	About 6.5 A40-hours
4	Run the Haiku cross-family reader on the existing 2,060 items	A stronger, different-family reader tests whether the data carries readable meaning; code already written	About $0.20 (cap $5)
5	Treat the verbal route as closed at 7B and spend effort on statistical/likelihood tools, with a validity check on the control pair built in	The verbal null is uninformative; the statistical route only fails on its control	none

What would change my mind: if the leave-cat-out test (item 1) shows the cat feature adds more than every other animal's feature, the dead end becomes a positive lead. If it does not, the data-side route is closed at this level and the useful object is a model-side detector.

6. Caveats and what is not claimed

• All detector numbers come from one deterministic run each; the intervals are bootstrap intervals over the examples.
• The teacher check missed its pre-registered threshold by one sample (89/100 contain "cat"; the other 11 said "Feline", "Purrfect", "Kitty"). Track A reports that as a setup failure, Track B (a word list fixed before any reader ran) as a pass. Both are reported; neither overrides the other.
• The published reference values are approximate readings of the paper's figure; the paper's own datasets are not in the repository, so every 'reproduction' regenerates the data.
• The decision rules for the detectors were written before the outputs, but a count of unigram frequencies was made first and is disclosed in the decision log; the neutral-pair gate was changed from one-sided to two-sided on 20 Oct (D9) before it was applied.
• Nothing here used an LLM judge or any API call.

Sources: results_RQ0.md; experiments_RQ0/outputs/results_RQ0.json; experiments_leak_test/results_leak_test_qwen_hpc.md and results_sensitivity_parse_qwen.md; experiments_RQ1/outputs_hpc/{e1d,e1e,e2}_results.json; experiment_plan_RQ1.md; experiment_plan_RQ2.md; experiments_leak_test/DECISIONS.md.